Problem 13 | Problems T5_MechanicalProperties.pdf #13 | Vickers · Brinell hardness

Vickers and Brinell Hardness Numbers from Indentation Measurements

Original statement · Problems T5_MechanicalProperties.pdf, p. 5

13. Calculate the Vickers and Brinell hardness numbers for a metallic material on which several hardness tests were performed, obtaining the following results:

- *a. When the diamond Vickers hardness test with a load of 30 kg was performed, the average length of the diagonal was: first impression 0.527 mm, second impression 0.481 mm, third impression 0.497 mm.*
- *b. When using the Brinell test with a ball of 10 mm and a load of 3000 kg, the diameter of the indentation was 4.01 mm.*

Official Solution: 220 HV, 227 HB

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Vickers.** The indenter is a diamond right pyramid with 136° between opposite faces; the hardness is the load divided by the lateral area of the impression, $HV = F/A_{imp}$, which depends only on the mean diagonal d [Session 12 Slide 26]. The slide defines d as the arithmetic mean of the two diagonals d_1 and d_2 of one impression; here the sheet gives one diagonal length for each of the three impressions, which are averaged first.
2. **Brinell.** The hardness is the load divided by the area S of the spherical cap left by the ball, $HB = P/S$ with P in kg [Session 12 Slide 19]. The result is a number with no units attached by convention.
3. The tests are performed under their standard conditions (flat surface, indentation far from edges and from other impressions) [Session 12 Slide 28].

Data

Quantity	Symbol	Value
Vickers load	F	30 kgf (HV30)
Mean diagonal, impression 1	d_1	0.527 mm
Mean diagonal, impression 2	d_2	0.481 mm
Mean diagonal, impression 3	d_3	0.497 mm
Brinell load	P	3000 kg
Ball diameter	D	10 mm

Quantity	Symbol	Value
Diameter of the indentation	D_i	4.01 mm

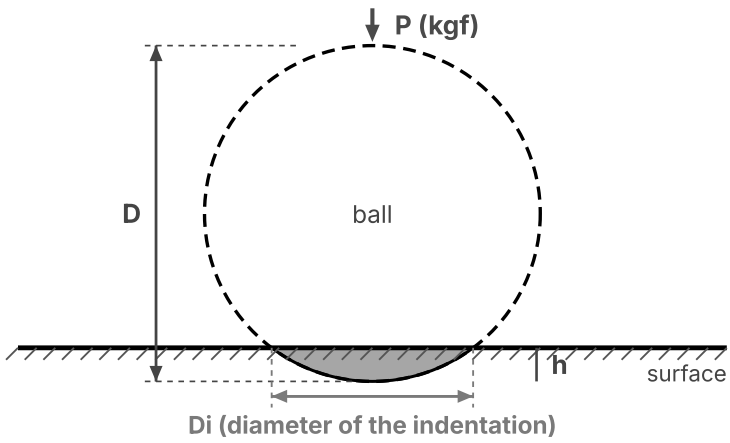
Unknowns: HV (a) and HB (b). Both loads are kilogram-force, the unit used by the slides; the hardness numbers are quoted without units (kgf/mm² in the definition, $1\ \text{kgf} = 9.81\ \text{N}$).

PHASE 2 · FRAMES & STRATEGY

No frames are involved. Each hardness comes from the load and one measured length, through the formulas of the slides.

Strategy.

- (a) Average the three diagonals, as the slide does with d_1 and d_2, and use $HV = 2F\sin(136^\circ/2)/d^2 = 1.854\,F/d^2$. The expression is derived from the pyramid: the faces make $136°$ with each other, so the lateral area is $A = d^2/(2\sin 68^\circ)$ and $HV = F/A$.
- (b) Use $HB = 2P/\left[\pi D\left(D - \sqrt{D^2 - D_i^2}\right)\right]$, which is P over the area of the cap, $S = \pi D h$ with $h = (D - \sqrt{D^2 - D_i^2})/2$ the depth of the indentation.



Cap area S = π D h with h = (D – √(D² – Di²)) / 2 ⇒ HB = P / S

Figure 11. Brinell test: the cap of height h left by the ball of diameter D (the proportions are exaggerated, not to scale).

PHASE 3 · STEP-BY-STEP DERIVATION

a) Vickers hardness

Mean diagonal.

$$\bar{d} = \frac{d_1 + d_2 + d_3}{3} = \frac{0.527 + 0.481 + 0.497}{3} = \frac{1.505}{3} = 0.50167 \text{ mm}$$

Constant of the pyramid. $\sin 68^\circ = 0.927184$, so

$$2 \sin \frac{136^\circ}{2} = 2 \sin 68^\circ = 1.8544$$

Hardness.

$$HV = \frac{2F \sin(136^\circ/2)}{\bar{d}^2} = \frac{1.8544 \times 30}{(0.5017)^2} = \frac{55.631}{0.251669} = \mathbf{221.0\ HV30}$$

so the hardness is about **221 HV**.

b) Brinell hardness

The indentation is characterised by its diameter $D_i = 4.01$ mm, with the ball of $D = 10$ mm:

$$D_i^2 = 4.01^2 = 16.0801 \qquad D^2 - D_i^2 = 100 - 16.0801 = 83.9199 \text{ mm}^2 \qquad \sqrt{D^2 - D_i^2} = 9.16078 \text{ mm}$$

$$D - \sqrt{D^2 - D_i^2} = 10 - 9.16078 = 0.83922 \text{ mm} \qquad \pi D \left(D - \sqrt{D^2 - D_i^2}\right) = \pi \times 10 \times 0.83922 = 26.3649 \text{ mm}^2$$

The depth of the indentation is $h = 0.83922/2 = 0.4196$ mm and the area of the cap $S = 26.3649/2 = 13.182$ mm². Therefore

$$HB = \frac{2P}{\pi D\,(D - \sqrt{D^2 - D_i^2})} = \frac{2 \times 3000}{26.3649} = \mathbf{227.6\ HB}$$

that is, about **228 HB** (equivalently $P/S = 3000/13.182 = 227.6$).

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** F in kgf and d in mm give kgf/mm², the unit of hardness; $D^2 - D_i^2$ in mm², and the Brinell denominator is an area (mm²), so P/S is again kgf/mm².
- **Load factor.** $Q = P/D^2 = 3000/10^2 = 30$, one of the standard values of the table of [Session 12 Slide 20] for the ball of 10 mm.
- **Order of magnitude.** $HV \approx 221$ and $HB \approx 228$ are of the same order, since both scales are defined as a load in kgf over an area in mm²; but the slides remind us that there is no universal conversion between methods [Session 12 Slide 15].
- **Limit.** If $D_i \to 0$ then $D - \sqrt{D^2 - D_i^2} \to D_i^2/(2D)$ and $HB \to 4P/(\pi D_i^2)$, the load over the area of the circle of the indentation, as it should for a very shallow cap.

Interpretation

The metal is a medium-hard alloy, of the order of 220 kgf/mm². The three Vickers impressions differ by 10 % in their diagonals (0.481 to 0.527 mm), which changes the hardness by about 20 % between impressions (240.5 to 200.3 HV): this is the scatter of the measurement, and the reason why several tests are averaged.

AVERAGE OF THE DIAGONALS, NOT OF THE HARDNESS NUMBERS

The slide defines d as the mean of the diagonals of the impression, so the diagonals are averaged first and one hardness is computed. Averaging the three hardness numbers instead (225.2, 240.5, 200.3) would give 222.0 HV, which is not the value defined by the slide.

Check against the official solution

Item	Official	Computed	Comment
Vickers	220 HV	221.0 HV	agrees to the rounding of the key
Brinell	227 HB	227.6 HB	agrees (the key truncates)

SMALL DIFFERENCE WITH THE OFFICIAL SOLUTION: VICKERS

The key gives 220 HV and the computation, with 1.854 from $2 \sin 68^\circ = 1.8544$, gives 221.0 HV. The difference is below 0.5 %. The computed value is kept; the key looks rounded down to the tens.